Curriculum Vitae

Mohammad Ful Hossain Seikh | fulhossain@ku.edu | Website

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Education

Ph.D. in Physics *2020–2026*
University of Kansas, USA **3.964/4.0**

M.S. in Electrical Engineering *2024–2025*
University of Kansas, USA **3.778/4.0**

M.Sc. in Physics *2018–2020*
Aligarh Muslim University, India **9.170/10** (Gold Medalist)

B.Sc. Honors in Physics *2015–2018*
Aligarh Muslim University, India **9.186/10** (Gold Medalist)

Research Experience

Research Roles and Appointments

- **Postdoctoral Researcher: Particle Astrophysics, University of Kansas** *Jun 2026–Present*
  Research on ultra-high energy particle detection with radio techniques, including the ARA five-station data analysis, radio-background characterization, and detector studies with Prof. D. Besson.
- **Core Analyst, ARA Five-Station Diffuse Neutrino Search** *2022–Present*
  One of eight analyzers conducting the first array-wide diffuse ultra-high energy neutrino search using 10.6 years of ARA data. Integrated the ARA01 analysis into the common multi-station framework and contributed to array-wide optimization, unblinding, and interpretation of the final result.
- **Analysis Lead, Non-Anthropogenic Down-going Signals in ARA** *Jul 2026–Present*
  Leading an array-wide search across all five ARA stations for down-going radio signals inconsistent with known anthropogenic backgrounds.
- **Analysis Lead, Radio Background Characterization with ARA** *Dec 2025–Present*
  Leading studies of astrophysical, environmental, and anthropogenic radio backgrounds in ARA data, including solar activity, galactic noise, high-wind triboelectric events, radiosondes, and local sources.
- **Primary Analyzer, Askaryan Radio Array (ARA) Station 1** *2021–2026*
  Led the end-to-end ARA01 analysis for ultra-high energy neutrino searches, including detector calibration and data-driven modeling, event reconstruction, background rejection, and simulation. Developed the station-specific analysis framework later incorporated into the full-array search.
- **Monitoring Shifter, IceCube Neutrino Observatory** *2021–2026*
  Served as KU representative for monitoring shifts, presenting weekly reports to the collaboration.
- **Visiting Summer Student** *May–Jul 2019*
  Studied "Importance of 1-3 Mixing Angle in Three-Flavor Neutrino Oscillations Paradigm" with Prof. S.K. Agarwalla at the Institute of Physics, India, producing technical report DOI: 10.13140/RG.2.2.34421.12009.
- **Winter Research Intern** *Dec 2019–Jan 2020*
  Researched "Weak Interaction Physics" at Indian Institute of Technology–BHU with Prof. G. Abbas.

Selected Dissertations, Theses, and Projects

- **Physics PhD Dissertation** *2020–2026*
  "Toward a Five-Station Diffuse Ultra-High Energy Neutrino Search with the Askaryan Radio Array: ARA01 Analysis Framework." Defended May 8, 2026. Advised by Prof. D. Besson. Published on ProQuest ISBN: 9798252480190 and KU ScholarWorks.
- **Electrical Engineering Master's Project** *2024–2025*
  Developed **AAFIYA**, an open-source Python toolkit for antenna measurement, simulation, and characterization, with Prof. Jim Stiles at KU EECS. The project led to an accepted contribution at the 2026 IEEE AP-S/URSI Symposium. (GitHub), (Report), (Slides).

- **Machine Learning Course Project** *2025*
  Developed a CNN model for "Automatic Surface Detection in Echograms using Deep Learning" in a KU EECS machine-learning course. (GitHub), (Report).
- **Physics Master's Thesis** *2019–2020*
  Investigated light sterile neutrino oscillations with Prof. M.S. Athar at Aligarh Muslim University, producing the review report "Light Sterile Neutrinos: Status and Prospects." DOI:10.13140/RG.2.2.13777.66405.
- **Physics Undergraduate Project** *2017–2018*
  "Quarks & Leptons": detailed study of Standard Model interactions with Prof. M.S. Athar at AMU (Handwritten Report).

Scientific Software and Open-Source Development

- **MAHAM: M**odules for **A**stroparticle, **H**igh-energy, **A**stronomy and **M**ultimessenger *2026–Present*
  Founder and lead developer of `Project MAHAM`, an open-source Python toolkit under active development for reusable, validated, and citable scientific methods and data products in astroparticle physics, high-energy astrophysics, astronomy, and multimessenger research. Established the package architecture, scientific provenance and validation framework, with current development spanning neutrino spectra and models, experimental limits and sensitivities, detector-response calculations, statistical methods, and reproducible literature datasets.
- **AAFIYA: A**ntenna **A**nalysis in **F**requency-domain for **I**mpedance & **Y**ield **A**ssessment *2024–Present*
  Creator and developer of `AAFIYA`, an open-source Python toolkit for reproducible antenna characterization using measurement and simulation data. Implements S-parameter and impedance analysis, gain and beam-pattern characterization, beamwidth, front-to-back and cross-polarization metrics, realized vector effective length, and measurement-to-simulation validation, with ongoing development toward a more integrated reusable software package.
- **AraProc: ARA Data Analysis Framework** *2024–Present*
  Co-developer of the ARA Collaboration's `AraProc` Python analysis framework, with 21 merged pull requests covering waveform analysis, event reconstruction, CW filtering, and data quality improvements.

Field Experience

- **NSF's Summit Station, Greenland Field Campaign** *Apr–May 2025*
  Participated in the Radio Neutrino Observatory in Greenland (RNO-G) calibration campaign, performing in situ radio-frequency measurements and equipment deployment at Summit Station, Greenland.

Publications & Manuscripts

Manuscripts in Collaboration Review

Core analysis team: Bishop, A., Clark, B.A., Dasgupta, P., Giri, P., Machtay, A., Muzio, M.S., Salcedo-Gomez, A., and **Seikh, M.F.H.**

- **ARA Collaboration** *(2026).* "First Array-Wide Search for Ultrahigh Energy Neutrinos with the Askaryan Radio Array." *Under Collaboration review; planned submission to Physical Review Letters.*
- **ARA Collaboration** *(2026).* "Search for Ultrahigh Energy Neutrinos using 10.6 Years of Askaryan Radio Array Data." *Under Collaboration review; planned submission to Physical Review D.*

Refereed Journal Publications (First/Lead Author)

- **Seikh, M.F.H.** *(2025).* "A geometrical approach to neutrino oscillation parameters". *AIP Advances 15 (10), 105204.* DOI:10.1063/5.0292233.
- **Seikh, M.F.H., Grachev, V., Kravchenko, I., Athar, M.S., & Besson, D.** *(2025).* "First Radio Neutrino & Cosmic Rays Astronomy Workshop in India". *Eur. Phys. J. Spec. Top. 234(16), 4919–4924.* arXiv:2509.20704; DOI:10.1140/epjs/s11734-025-01955-8.
- **Seikh, M.F.H.** *(2025).* "Physics of radio antennas". *Eur. Phys. J. Spec. Top. 234(16), 4967–4983.* arXiv:2411.07507; DOI:10.1140/epjs/s11734-025-01500-7.
- **Seikh, M.F.H. for the ARA Collaboration** *(2025).* "Askaryan Radio Array: searching for the highest energy neutrinos". *Eur. Phys. J. Spec. Top. 234(16), 5065–5075.* arXiv:2411.01761; DOI: 10.1140/epjs/s11734-025-01708-7.

Conference Proceedings (First/Lead Author)

- **Seikh, M.F.H., Jarvis, R., & Stiles, J.** *(2026).* "AAFIYA: Antenna Analysis in Frequency-domain for Impedance and Yield Assessment". *IEEE Xplore, pp. 2537-2540.* DOI:10.1109/AP-S/USNC-URSI60190.2026.11675620; arXiv:2601.13583.
- **Seikh, M.F.H., Giri, P. & Besson, D. for the ARA Collaboration** *(2025).* "Techniques for Continuous Wave Identification and Filtering in the Askaryan Radio Array". PoS(ICRC2025)1171; arXiv: 2509.20735.
- **Seikh, M.F.H. & Besson, D. for the RNO-G Collaboration** *(2025).* "Observation of Radio Solar Flares in RNO-G". PoS(ICRC2025)1365.
- **Seikh, M.F.H. for the ARA Collaboration** *(2023).* "Calibration and Physics with ARA Station 1". PoS(ICRC2023)1163; arXiv:2308.07292.

Invited and General Scientific Writing

- **Seikh, M.F.H. & Besson, D.** *(2026).* "Polar Ice as a Laboratory for Physics". *Physics Bulletin 2026, Physics Department, AMU*, pp. 27–37.
- **Seikh, M.F.H. & Besson, D.** *(2024).* "IceCube Neutrino Observatory: A Cold Discovery Hub". *Physics Bulletin 2024, Physics Department, AMU*, pp. 14–23.
- **Seikh, M.F.H.** *(2020).* "Neutrinos and the 40 Future Routes". *Physics Bulletin, Physics Department, AMU.* DOI:10.13140/RG.2.2.27981.10729.

Collaborative Works

Complete publication record, including contributions to the ARA, RNO-G, ARIANNA, IceCube, IceCube-Gen2, PUEO, TAROGE-M, and RET Collaborations, available on:

Inspire HEP ; Google Scholar ; ResearchGate ; ORCiD ; Scopus

Invited Lectures, Seminars, & Colloquia

- **The Search for the Highest-Energy Neutrinos in Polar Ice**
  Eddy and April Lucas Physics Seminar, Wichita State University *(Oct 21, 2026, scheduled)*
- **HEP Seminar**, Oklahoma State University *(Nov 5, 2026, scheduled)*
- **Neutrinos in Multimessenger Astronomy**
  Guest Lectures, ASTR 798 High-Energy Astrophysics, University of Kansas *(Dec 1 & 3, 2026, scheduled)*
- **WUCAP Seminar**, Washington University in St. Louis *(Feb 12, 2027, scheduled)*
- **Physics Colloquium**, Missouri University of Science and Technology *(Spring 2027, date TBD)*
- **Searching for the Universe's Highest-Energy Neutrinos with Radio Detectors**
  Particle Physics Seminar, Carnegie Mellon University *(Sep 9, 2026)*
- **Radio Engineering for Particle Astrophysics Experiments**
  HEP Seminar, University of Nebraska-Lincoln *(Mar 26, 2025)*
- **Radio Engineering for Particle Astrophysics Experiments**
  Astroparticle Seminar, Ohio State University *(Jan 22, 2025)*
- **Radio Engineering & Antenna Physics**
  Lecture Series, IEI Summer School 2024, University of Alaska, Anchorage *(Jul 2–5, 2024)*
- **UHE Neutrino & Cosmic Ray Detectors**
  ICEHAP Seminar, Chiba University, Japan *(Aug 9, 2023)*
- **Antenna Physics & Measurements**
  Lecture Series, IEI Summer School 2023, South Dakota School of Mines *(Jul 2–5, 2023)*
- **UHE Neutrino & Cosmic Ray Detectors**
  Physics Colloquium, AMU, India *(Jan 12, 2023)*
- **Antennas in UHE Neutrino Detectors**
  Particle Physics Seminar, University of Delaware *(Oct 4, 2022)*

Teaching & Research Mentoring

Teaching & Tutoring

- **PHSX 211: General Physics I, KU** *Fall 2021*
  Course Instructor and Discussion Moderator. Classical mechanics and thermodynamics for engineering and physical science majors.
- **PHSX 115: College Physics II, KU** *Fall 2021*
  Lab Instructor. Electricity, magnetism, optics, and introductory atomic and nuclear physics.
- **PHSX 114: College Physics I, KU** *Summer 2021*
  Lab Instructor. Mechanics, fluids, heat, thermodynamics, and sound.
- **Private Tutor, Aligarh, India** *2016–2020*
  Taught high school physics and mathematics during B.Sc. and M.Sc. studies.

Research Advising

- **Daniel Dakroub**, Senior Undergraduate, MEPhI, Moscow, Russia *Apr 2026–Present*
  Project: Inter-Experiment Search for Surface-Mode Radio Propagation.
- **Burhan-Ullah Khan**, High School Student, RMPS AMU City School, Aligarh, India *Sep 2026–Present*
  Project: Model Development for Explaining ANITA Anomalous Events.

Professional Service

Editorial

- **Book Editor** for Springer's EPJ-ST journal: "Radio Detection of Ultra-high Energy Neutrino & Cosmic Ray" (15 articles) with M.S. Athar, D. Besson, V. Grachev, and I. Kravchenko (View Cover) *(2024–2025)*
- **Founder and Editor-in-Chief** of *The Magnetic Monopole* e-magazine for KU's Society of Physics Students (View Issue) *(Jun 2022)*

Research Article Review

- **Peer Reviewer:** for World Scientific's IJMP-A journal (1 article, 2 reviews) (2025)
- **Peer Reviewer:** for Springer's EPJ-ST journal (5 articles, 11 reviews) *(2024–2025)*
- **Internal Collaboration Reviewer:**
  - ✓ Muzio, M.S. and Salcedo-Gomez, A. for the ARA Collaboration, "The First Array-Wide Diffuse Flux Search for UHE Neutrinos with the Askaryan Radio Array", ARENA2026, *(Jun 2026)*
  - ✓ Giri, P. for the ARA Collaboration, "A Comprehensive Diffuse Neutrino Search Using the Full Askaryan Radio Array", Lepton–Photon2025, *(Nov 2025)*

Conference Session Chair

- **General Cosmology:** for APS Global Physics Summit (Mar 19, 2026)
- **Topical Groups Session II:** for APS Global Physics Summit (Mar 17, 2026)

Other Reviews

- **REU Application Reviewer:** for KU Physics and Astronomy (10 Applications) (2025)
- **SSGSA Application Reviewer:** for SSGSA (70+ Applications) *(2021–2026)*

Professional Leadership & Membership

- **American Physical Society (APS)**: Member *(2020–Present)*
- **National Society of Black Physicists**: Member *(2021–Present)*
- **Physics & Astronomy Graduate Student Organization (GSO), KU**:
  - ✓ Graduate Students Representative - Member at Large *(2025–2026)*
  - ✓ Graduate Students Representative - Liaison *(2024–2025)*
  - ✓ GSO Executive Officer *(2023–2024)*
- **Sir Syed Global Scholar Award (SSGSA)**:
  - ✓ Peer Mentor *(2023–Present)*
  - ✓ Team Lead, Promotions & Outreach *(2023–2025)*

- **Society of Physics Students, KU Chapter**: Graduate President *(2021–2022)*
- **Sigma Pi Sigma (National Physics Honor Society)**: Lifetime Member
- **AMU Society of Physics**: Member *(2018–Present)*

Technical Proficiencies

- **Programming**: Python, C/C++, Fortran
- **Scientific Computing & Data Analysis**: CERN ROOT, uproot, NumPy, SciPy, pandas, scikit-learn
- **Detector, Simulation & RF Software**: AraRoot/AraProc, AraSim, NuRadioMC, HFSS, Mathematica
- **Development & Computing**: Git/GitHub, Linux (Ubuntu), macOS, Windows
- **Document Preparation**: LaTeX, Microsoft Office Suite

Honors & Awards

Academic Honors

- **University Gold Medals**:
  - ✓ M.Sc. Physics (Top among 50 students), AMU *(2020)*
  - ✓ B.Sc. Honors Physics (Top among 120 students), AMU *(2018)*
  - ✓ Highest percentage in B.Sc. (Hons) Faculty of Science, AMU *(2018)*
- **Abdul Aziz Medal** for B.Sc. Physics, AMU *(2018)*
- **Undergraduate Topper Scholarship**, UGC India *(2020)*

Fellowships & Grants

- **Ralston Dissertation Fellowship**, KU Physics *(2025–2026)*
- **McKeithan Summer Fellowship**, KU Physics *(2025)*
- **Graduate Student Award for Distinguished Service**, KU *(2024–2025)*
- **Wallace S. Strobel Scholarship**, KU EECS *(2024–2025)*
- **Ralston Summer Fellowship**, KU Physics *(2024)*
- **Doctoral Student Research Fund**, KU *(2024)*
- **Distinguished Students Award**, APS FIP *(2021–2022)*
- **Carl & Catherine Chaffee Scholarship**, KU Physics *(2020–2021)*
- **University Merit Financial Award**, AMU *(2019–2020)*

Conference Travel Awards

- **Junior Participant Support**, Neutrino *(2026)*
- **Graduate Student Travel Fund**, KU Physics *(2023, 2025)*
- **DAP Travel Grant**, APS *(2024)*
- **DPF Travel Award**, APS *(2022, 2024)*
- **CLAS Graduate Scholarly Development Fund**, KU *(2022)*

Other Distinctions

- **Washington DC Scholarship**, AMU Alumni Association *(2015–2019)*
- **Sir Syed Global Scholar Award** for prospective graduate students *(2018)*
- **Senior Diploma in Painting with Distinction**, Bangiya Sangeet Parishad (2014)

Conference Presentations

Oral Presentations

- **Pioneering an Array-Wide Search for Ultra-high Energy Neutrinos with ARA**
  APS Global Physics Summit *(Mar 18, 2026)*
- **RNO-G in-ice Antennas**
  RNO-G Collaboration Meeting, Uppsala University *(Sep 9–14, 2024)*
- **Radio Frequency Instrumentation**
  Radio Neutrino Workshop, AMU *(Apr 18–20, 2024)*

- **Towards Five Station ARA Analysis**
  APS April Meeting, Sacramento *(Apr 3–6, 2024)*
- **ARA Antenna Models**
  ARA Collaboration Meeting, Ohio State University *(Mar 5–8, 2024)*
- **Current Status & Future of ARA**
  APS Prairie Section Meeting, University of Missouri *(Nov 30–Dec 2, 2023)*
- **Introduction to Antenna Physics**
  RET Collaboration Meeting, Vrije University Brussels *(Nov 8–11, 2023)*
- **Gain Calibration of RNO-G Antennas**
  RNO-G Fall Meeting, Penn State University *(Sep 11–15, 2023)*
- **Galactic Noise in Radio Neutrino Experiments**
  Astro Seminar, KU *(Oct 28, 2022)*
- **Towards ARA Station 1 Calibration**
  APS April Meeting, New York City *(Apr 9–12, 2022)*
- **RNO-G Ice Calibration Planning**
  RNO-G Collaboration Meeting, University of Chicago *(Apr 8, 2022)*

Poster Presentations

- **A Diffuse Search for UHE Cosmogenic Neutrinos with ARA Five Stations**
  Neutrino 2026, UC Irvine *(Jun 21–26, 2026)*
- **Monitoring Snow Accumulation in ARA Using DAQ Temperature Data**
  Neutrino 2026, UC Irvine *(Jun 21–26, 2026)*
- **A Geometrical Approach to Neutrino Oscillation Parameters**
  APS Global Physics Summit *(Mar 17, 2026)*
- **CW Identification & Filtering Techniques in ARA**
  39th ICRC, Geneva, Switzerland *(Jul 19, 2025)*
- **Observation of radio solar flares in RNO-G**
  39th ICRC, Geneva, Switzerland *(Jul 15, 2025)*
- **Gain Measurements of LPDA Antennas**
  APS April Meeting, Sacramento *(Apr 3–6, 2024)*
- **Overview of ARA Detector: Five Station Analysis**
  ICTS Neutrino School, India *(Apr 22–May 3, 2024)*
- **Calibration of ARA Station 1**
  38th ICRC, Nagoya University, Japan *(Jul 29, 2023)*

Outreach & Leadership

- Co-organized and hosted India's inaugural workshop on **Radio Detection of UHE Neutrino and Cosmic Rays** at AMU *(Apr 2024)*
- Conducted outreach seminar on **Physics Research Opportunities** at AMU *(2021)*
- Participated in India's first **Science Leadership Workshop** *(2020)*